

Q2 Give the design choices for below scenarios and their explanation.



HEALTHCARE

Clinical Intake Drug-Interaction Flag

A hospital's intake system needs to flag potential multi-drug interaction risks as a nurse enters a new patient's medication list — a single-patient, moderate-urgency check during intake, not a batch job. It must run inside the hospital's own network per data-governance policy, and getting the interaction reasoning right matters for patient safety.

DESIGN CHOICE 1 — MODEL FAMILY

Foundation LLM

Small Language Model (SLM)

Reasoning Model

DESIGN CHOICE 2 — WHERE IT RUNS

Cloud

Edge / On-Prem

For this scenario I would like to go with the below explanation:

Model : **Reasoning Model**

Explanation: **As the hospital system needs to know why a flag was raised and understand the concern. Reasoning Model can provide clear, step-by-step rationale and efficient enough.**

Where to Run: **Edge/ On-Prem**

Explanation: **As it is patient related information and more sensitive, so it needs On-Prem environment for reliability, real time checking during manual patient intake without relying on the external internet connectivity.**



GOVERNMENT / DEFENSE

Sovereign Case-File Summarization

A government agency needs to summarize case files on an air-gapped network — no connection to any public cloud is permitted, for any request, ever. Case files are lengthy and require genuine multi-step synthesis (cross-referencing dates, names, and prior rulings), not just extraction.

DESIGN CHOICE 1 — MODEL FAMILY

Foundation LLM

Small Language Model (SLM)

Reasoning Model

DESIGN CHOICE 2 — WHERE IT RUNS

Cloud

Edge / On-Prem

For this scenario I would like to go with the below explanation:

Model : Reasoning Model

Explanation: As Case files are explicitly noted as lengthy and complex, requiring cross-referencing across timeline dates, entities (names), and legal precedents (prior rulings), it would much better to use Reasoning Model as it uses chain-of-thought to systematically chart the relationships across different parts of the document, ensuring the final text is genuine.

Where to Run: **Edge/ On-Prem**

Explanation: **Edge / On-Prem** allows the entire model architecture, weights, and context windows to operate purely within isolated local servers. This completely eliminates any threat of data exfiltration or external surveillance, aligning perfectly with the agency's strict security protocols.